UNIT 5 — HEAT

Revision notes based on the supplied textbook pages, with key definitions, formulas, comparisons and applications.

1. HEAT AND TEMPERATURE

• **Heat** is energy transferred from a hotter body to a colder body because of a temperature difference. Heat flows naturally from higher temperature to lower temperature until thermal equilibrium is reached.

• The SI unit of heat energy is the **joule (J)**. Another commonly used unit is the calorie. 1 calorie ≈ 4.184 joules.

• A sensation of hotness or coldness is not a reliable measurement of temperature; a thermometer is used to measure temperature.

• **Temperature** indicates how hot or cold a body is and is related to the average kinetic energy of its particles. Heat is energy in transfer; temperature is a measure of thermal state.

2. TEMPERATURE SCALES

Scale	Symbol	Lower fixed point	Upper fixed point	Divisions
Celsius	°C	0°C	100°C	100
Fahrenheit	°F	32°F	212°F	180
Kelvin	K	273.15 K	373.15 K	100

Conversion relation: C/5 = (F – 32)/9 = (K – 273.15)/5

• Kelvin is the SI unit of thermodynamic temperature. Write kelvin as K, not °K.

• 0°C = 32°F = 273.15 K; 100°C = 212°F = 373.15 K.

• To convert Celsius to Fahrenheit: F = (9/5 × C) + 32. To convert Celsius to Kelvin: K = C + 273.15.

3. EVAPORATION AND BOILING

Evaporation	Boiling
Occurs at the surface of a liquid.	Occurs throughout the liquid.
Can occur at a range of temperatures.	Occurs at a particular boiling temperature for a given pressure.
Usually a slow, surface process.	Rapid formation of vapour bubbles throughout the liquid.
Produces cooling because higher-energy particles escape.	During boiling at constant pressure, temperature remains constant while the liquid changes state.

• Evaporation can occur below the boiling point. Its rate increases with higher temperature, larger exposed surface area, greater air movement and lower humidity.

• Evaporation causes cooling: the faster, higher-energy particles escape, lowering the average kinetic energy of the remaining liquid.

• Boiling occurs when the vapour pressure of the liquid equals the external pressure. The boiling point changes with pressure.

4. THERMAL EXPANSION

Thermal expansion is the increase in dimensions of a substance when its temperature rises. Most materials contract on cooling.

Type	What increases?	Example
Linear expansion	Length	A heated metal rod becomes longer.
Area expansion	Length and breadth; surface area increases	A heated metal sheet expands in area.
Volume expansion	Length, breadth and height; volume increases	A heated metal block expands in volume.

• For most substances, gases expand more than liquids, and liquids expand more than solids for the same temperature rise (for ordinary conditions).

Examples and applications of expansion

• **Glass cracking:** glass is a poor conductor. Uneven heating makes one part expand before another, producing stress that may crack the glass.

• **Railway tracks and bridges:** expansion gaps or joints allow materials to expand in hot weather without buckling.

• **Pendulum clocks:** a metal pendulum rod expands in summer, increasing its length and the time period; the clock tends to lose time. In winter it contracts and the clock tends to gain time. Invar has a very small expansion coefficient.

• **Metal screw cap:** warming the metal cap makes it expand and can loosen it from the bottle neck.

5. TRANSFER OF HEAT

Heat is transferred by three main methods: conduction, convection and radiation.

Method	How it occurs	Where it occurs
Conduction	Transfer through neighbouring particles by interactions/collisions; there is no bulk movement of the material.	Mainly solids; also occurs in fluids.
Convection	Transfer by bulk movement of a fluid. Warmer, less-dense fluid generally rises while cooler, denser fluid sinks.	Liquids and gases.
Radiation	Transfer by electromagnetic waves; no material medium is required.	Can travel through vacuum.

• **Conduction example:** one end of a metal spoon becomes hot when the other end is placed in hot water.

• **Convection example:** water circulates as it is heated in a vessel. Warm water rises and cooler water moves down to replace it.

• **Radiation example:** heat from the Sun reaches Earth through space. Radiant heat travels at the speed of light.

6. LAND AND SEA BREEZES

• **Sea breeze (day):** land heats faster than the sea. Air over land becomes warmer and rises; cooler air from the sea moves towards land.

• **Land breeze (night):** land cools faster than the sea. Air over the relatively warmer sea rises; cooler air from land moves towards the sea.

• Both are caused mainly by convection due to unequal heating of land and water.

7. GOOD AND BAD CONDUCTORS OF HEAT

Good conductors	Poor conductors / insulators
Allow heat to pass through easily.	Do not allow heat to pass through easily.
Most metals, e.g. copper, aluminium, iron and mercury.	Wood, glass, wool, plastic, mud, air and many gases.

Applications of good conductors

• Copper tubing is used in automobile radiators and heat exchangers because copper transfers heat readily.

• Cooling coils in refrigerators and air conditioners are made from metals that conduct heat effectively.

• Metal cooking vessels transfer heat to food efficiently.

Applications of poor conductors

• Wooden or plastic handles on cooking utensils reduce heat transfer to the hand.

• Woollen clothes keep us warm because fibres trap air; trapped air is a poor conductor.

• Polar animals' thick fur traps air and reduces heat loss.

• Birds fluff their feathers to trap insulating air between them and reduce heat loss.

• Ice may be covered with sawdust or gunny bags because trapped air slows heat transfer.

• Mud houses remain relatively cool in summer and warm in winter because mud is a comparatively poor conductor of heat.

8. IMPORTANT REASONING AND EXAM POINTS

• A larger exposed surface area generally increases the rate of evaporation and can increase the rate of cooling.

• A hot object cools faster when the temperature difference between it and the surroundings is larger, other conditions being similar.

• A metal bar held under a running hot-water tap is heated mainly by conduction from the hot water at the surface into the metal.

• Convection cannot occur in a vacuum because it requires bulk movement of a fluid.

• Radiation does not require air, water or any other material medium.

• A shiny, polished surface is generally a poor absorber and emitter of thermal radiation compared with a dull black surface.

9. QUICK FORMULA REVISION

Quantity	Formula / value
Celsius–Fahrenheit	$F = (9/5)C + 32$
Fahrenheit–Celsius	$C = (5/9)(F - 32)$
Celsius–Kelvin	$K = C + 273.15$
Kelvin–Celsius	$C = K - 273.15$
Fixed points	0°C = 32°F = 273.15 K; 100°C = 212°F = 373.15 K
Heat unit	1 calorie ≈ 4.184 J

Revise the definitions, temperature conversions, differences between evaporation and boiling, examples of expansion, and the three methods of heat transfer.